

A Methodological Framework for Identifying and Empirically Validating Macroeconomic Risks Suitable for On-Chain Convex Hedge Instruments

Record of Closed Iterations and Request for Supervisor Guidance

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2026-05-06

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§1. Introduction & Economic Motivation

This memorandum is an empirical-methodology record and a request for guidelines. It documents the iteration history that the analytics workstream has executed to date, presents the methodology that produced each verdict, and asks five prioritised methodological questions whose answers we expect to bind the design of subsequent iterations. The memorandum is addressed to the analytics supervisor; we expect the supervisor's response, where given, to serve as the design constraint for the next iteration's pre-registration.

The project's economic motivation is best stated as a financial-intermediation gap for an identifiable underserved population rather than as an abstract claim about aggregate inequality. The population the iteration roster currently centres on is Latin-American digital workers paid in local currency (Colombian pesos, in the dev-AI Stage-1 iteration of Section 7) who carry a recurring USD-denominated cost burden through their professional inputs — subscription AI services, application programming interface usage fees, cloud compute, and adjacent software-as-a-service spend. These workers are exposed to FX-pass-through cost shocks on their monthly USD spend that no Colombian retail-bank product currently hedges. Conventional FX-derivative venues are inaccessible: minimum notional sizes, counterparty credit standing, and intermediating-broker requirements are calibrated for corporate treasury clients, not for an individual digital worker carrying a hundred-to-low-thousands USD monthly cost line. The intermediation gap is the operative phenomenon; aggregate distributional inequality is, under this reading, the cumulative consequence of many such uninsured intermediation gaps rather than the framing primitive.

We treat the framework's contribution as the identification and empirical validation of the macro risks X that produce the cost shocks, followed by the design of *parametric cost-insurance products* that close the intermediation gap. This framing is deliberate: it places the methodology under the standards of the parametric-index-insurance literature (Carter et al., 2017; Mahul and Stutley, 2010; Clarke, 2016) rather than under the standards of generic derivatives pricing, and it lets the supervisor evaluate the empirical work using the underwriting discipline he would apply to any insurance product (basis risk, adverse selection, catastrophic-loss probability, counterparty matching). The implementation venue described in Section 4 is on-chain settlement infrastructure; that infrastructure is not the economic primitive.

The wage-to-capital boundary remains a documented empirical phenomenon and is preserved as the analytical object of the iteration roster. Mendieta-Muñoz (2017a) documents premature deindustrialization in Colombia and the corresponding reallocation of labor away from productivity-rising sectors; Rodrik (2016b) establishes the cross-country pattern; McMillan and Rodrik (2011b) formalises the structural-change-and-productivity decomposition that underlies the pattern. Together these references establish that for a substantial subset of developing-economy populations, the path from wage income into productive-capital ownership has become structurally narrower over the past three decades, and that the populations on the wrong side of that narrowing are exactly those that an absent intermediation layer fails to insure. The project's empirical question is which macro risks X are quantitatively responsible for the recurring uninsured cost shocks that keep specific target populations on the wrong side of this

boundary.

The unit of work is a triple (Y, M, X) defined formally in Section 2 and motivated by a hedge-target proposition in Section 3. The validation strategy estimates a composite slope $\hat{\beta}_{\text{composite}}$ of Y on lagged X , with sign expectation, lag structure, primary specification, and verdict thresholds pre-registered before the panel is touched. Section 5 (drafted by the data-methodology agent) presents the methodology in full; we mention it here only to fix the object the supervisor is being asked to review.

Iteration count and the cross-iteration multiple-comparison posture. As of the date of this memorandum, the project has run five iterations and *all five are closed or parked*: no Stage-1 work is in progress. The five are: one PASS verdict (Pair D, broad-services Section-G–T employment share against COP/USD lag, closed 2026-04-28); one closed FAIL on the pre-registered weekly primary with two pre-registered sensitivities producing positive 90 % confidence intervals (FX-vol-on-CPI-surprise, closed 2026-04-19); one closed EXIT_NON_REMITTANCE (Phase-A.0 remittance corridor, closed 2026-04-24); one parked-for-record (P1 Bittensor SN18 event-study apparatus); and — closed within the past week, on 2026-05-06 — one *sign-flipped* FAIL (dev-AI Stage-1, Colombian young-worker Section-J narrow-ICT employment share against COP/USD lag), in which the pre-registered positive-sign expectation was rejected at conventional significance with $\hat{\beta} = -0.146$. The dev-AI Stage-1 iteration also produced, on a pre-specified Section-M sensitivity arm (professional/scientific/technical/admin services, CIU Rev. 4 Sections {69–75}), a strong positive estimate $\hat{\beta} = +0.455$ at $p \approx 1.13 \times 10^{-6}$. That sensitivity result is consistent with — but does not formally resolve — a compositional reading of the Pair D PASS under which the headline Section-G–T positive estimate reflected Section-M-style transmission averaged across the broader G–T aggregate rather than a Section-J ICT-narrow signal. The Pair D Section-J/Section-M decomposition is flagged here for reference and is not litigated in this section; the methodological consequences — both for the Pair D PASS interpretation and for the cross-iteration multiple-comparison posture — are taken up in Sections 6 and 7.5.

Each individual iteration was pre-registered, and the within-iteration anti-fishing discipline is documented in Section 5. The *cross-iteration* multiple-comparison structure, however, is not addressed by within-iteration pre-registration: which (Y, X) pairs entered the iteration roster in the first place, which were dropped after EXIT, and which were deepened after PASS are decisions made by the project team, not by a pre-registered iteration-allocation rule. We acknowledge that this constitutes an open methodological question and we treat it explicitly in Section 7 (specifically Section 7.5) rather than presenting the five verdicts as if they were drawn from independent, pre-iteration-selected experiments. The supervisor’s guidance on the appropriate cross-iteration familywise-error or selection-bias mitigation regime is one of the five asks in Section 7.

The remainder of the memorandum is organised as follows. Section 2 defines the operating framework and the (Y, M, X) triple. Section 3 introduces the inequality-differential hedge proposition that motivates the iteration design and reads it as the underwriting target of a parametric cost-insurance product. The on-chain settlement substrate is summarised by a separate agent in Section 4 for completeness; it is contextual rather than evidentiary, and the supervisor need not

engage with it to evaluate the econometrics. Sections 5 and 6 present the methodology and the iteration log. Section 7 states the five methodological questions in priority order. Section 8 closes with the request for guidelines.

§2. Operating Framework: The (Y, M, X) Triple

This section defines the unit of analytical work that the iteration roster operates on. We adopt the same definitions used internally throughout the project’s specs, plans, and memory artifacts so that the supervisor’s comments map directly onto the documents that govern subsequent iterations.

§2.1 Definitions

The triple (Y, M, X) consists of:

- Y — the welfare-relevant outcome variable on which a target population’s exposure to the wage-to-capital transition is measured. Y is operationalised per population. Examples used in closed and active iterations include the Section J information-and-communication employment share for young Colombian workers, the realised volatility of a household consumption basket, and the cross-sectional differential between productive-capital returns and a wage-indexed consumption basket.
- X — the macro risk hypothesised to block the wage-to-capital transition for the target population. X identification is empirical and population-specific. The first-cut iteration question is always: which macro variable, at which lag, currently kills wage earners’ attempts to enter productive-capital ownership in this population? X enters the regression as a lag-distributed right-hand-side variable; the lag structure is pinned in the spec text before the panel is constructed.
- M — the settlement geometry of the candidate hedging instrument. M is introduced for completeness; it is declared out of scope for empirical validation and deferred to Stage 2 of each iteration. The on-chain operational details that determine the feasibility of M for a given (Y, X) pair are surveyed in Section 4 for the supervisor’s reference; the rest of the memorandum does not require familiarity with them.

§2.2 Stage discipline

Each iteration proceeds through three stages with explicit exit criteria:

1. **Empirical risk validation.** Estimate $\hat{\beta}_{\text{composite}}$ of Y on lagged X under a pre-registered specification. Exit on a positive- $\hat{\beta}$ verdict at the pre-committed significance threshold. We treat this stage as the object of the supervisor’s review.
2. **Ideal-scenario M sketch.** Conditional on a Stage-1 PASS, propose an instrument geometry on the on-chain substrate that *would* settle the empirical $\hat{\beta}_{\text{composite}}$ if deployed. Stage 2

does not require live counterparty capital. Exit on a settlement-feasible construction that hedges the validated correlation.

3. **Live deployment.** Conditional on Stage 2 and live counterparty capital, deploy and run an execution test. This stage is out of scope for the present memorandum.

The stage discipline is enforced by an internal anti-fishing rule: stage drift, in particular allowing Stage 2 instrument-design considerations to re-enter the Stage-1 specification choice, is treated as a silent design adjustment under the project's pre-registration discipline (see Section 5 and Section 7.4). The empirical exercises the supervisor is being asked to review are entirely Stage-1.

§2.3 Iteration order

The default iteration order is target-population dominant. We fix the target population first; we fix Y on a candidate inequality- or wage-to-capital-transition exposure measure for that population; we enumerate X candidates from the empirical risk surface that the population faces; for each surviving X we search the on-chain-eligible M space for tradability. The pair (Y, X) only graduates to instrument design once a settlement-feasible M exists. A closed iteration with verdict FAIL or EXIT informs the X -search prior for the next population. We do *not* re-run the same (Y, X) at adjusted thresholds after a FAIL; the within-iteration anti-fishing invariants carry forward to subsequent iteration design.

§2.4 Acknowledged limitations

Two limitations of the framework are load-bearing for the supervisor's review and worth stating explicitly in this section rather than in Section 7.

First, the iteration order described above does not, by itself, address cross-iteration multiple-comparison concerns. Iteration selection (which populations and which X candidates enter the roster) is a project-level decision that is not pre-registered in the same way that within-iteration specifications are pre-registered. We treat this as an open methodological question (Section 7.5).

Second, identification at Stage 1 is correlation-identified rather than causally identified. The project's PASS-verdict iteration (Pair D) explicitly hedges the correlation rather than asserting a causal channel (, per the closure memo `memory/project_pair_d_phase2_pass.md`, Reality-Checker FLAG #1). We discuss the identification posture and its implications for moving from Stage 1 to Stage 2 in Section 7.1.

§3. The Abrigo Lemma: Inequality Differential as Hedge Target

This section states the proposition that motivates the choice of Y in the framework's medium-term iteration roster. It is presented as a proposition rather than a proven theorem; the disclaimer is repeated immediately after the formal statement and is binding on the rest of the memorandum. The economic interpretation given to the proposition is that of a parametric-cost-insurance underwriting target rather than that of a generic derivative hedge target; the mathematical content is unchanged from earlier statements internal to the project, but the framing under which

the supervisor is invited to evaluate it is the parametric-index-insurance literature ([Carter et al., 2017](#); [Mahul and Stutley, 2010](#); [Clarke, 2016](#)).

Lemma 1 (Inequality-Differential Hedge Target). *Let $R_a(t)$ denote the return process of an asset basket representative of capital-holding households' wealth, and let $R_c(t)$ denote the return process of a consumption basket representative of working-class households' purchasing power. Define the inequality-differential process*

$$Y_{inequality}(t) = R_a(t) - R_c(t).$$

A hedging instrument settled on $Y_{inequality}(t)$ is welfare-relevant for working-class wealth in a way that no instrument settled on $R_a(t)$ alone or $R_c(t)$ alone is, because the differential directly indexes the relative-wealth gap that determines the working-class household's purchasing-power decline relative to capital-holding households.

Status of the Lemma — load-bearing disclaimer. Lemma 1 is stated as a proposition motivating the project's iteration design. It is *not* a proven theorem. In particular, the existence of a settlement-feasible instrument that materialises the differential $Y_{inequality}$ on the on-chain substrate, and the calibration of the associated stochastic process, are Stage-2 / Stage-3 work that this memorandum does *not* prove. The proposition's role in the memorandum is solely to motivate why the iteration roster targets distributional differentials rather than single-aggregate macro variables.

Insurance-framing reading. Under the framing developed in Section 1, the differential $Y_{inequality}(t)$ is the welfare-relevant index variable on which a parametric cost-insurance product should settle. The working-class household holds the policy — the *insured* party — and the policy pays out as a function of the realised differential. The capital-holding counterparty already carries the asset exposure represented by $R_a(t)$ through pre-existing portfolio positions and therefore does not need to be matched to a new opt-in derivative leg; the counterparty matching is structural rather than negotiated, in the sense documented for parametric-index insurance by [Carter et al. \(2017\)](#) and surveyed for developing-country agricultural applications in [Mahul and Stutley \(2010\)](#). The basis-risk posture that [Clarke \(2016\)](#) formalises — the gap between an index-payoff and the policy-holder's realised loss — becomes the appropriate Stage-1 evaluation discipline, and that posture is the one under which the supervisor is asked to evaluate the empirical work in Section 7.2.

§3.1 Why a single-aggregate hedge is insufficient

A standard single-aggregate macro hedge — for example, an instrument settled on the headline consumer-price index alone, or on the bilateral exchange rate alone — fails to capture the relative-wealth dimension that matters for working-class welfare. The headline consumer-price index mixes consumption baskets across the wealth distribution; capital-holding households frequently hold consumer-price-index-indexed assets that neutralise their own exposure to inflation, leaving the working-class household systematically more exposed than the headline number suggests. A

hedge settled on the headline index therefore prices off the *average* household's exposure rather than off the *relative* exposure differential that is welfare-relevant for the population of interest. Read in insurance terms, the headline-index policy carries unmodelled basis risk against the policy-holder's actual loss exposure; [Clarke \(2016\)](#) establishes that uninternalised basis risk on an index-payoff product is a first-order determinant of rational demand for that product. The differential $Y_{\text{inequality}}(t) = R_a(t) - R_c(t)$ removes the aggregation problem by construction: it indexes precisely the gap between the two baskets rather than the level of either, and therefore admits a basis-risk treatment of the kind [Clarke \(2016\)](#) requires.

§3.2 Two-sided market and counterparty matching

A second motivation for the differential formulation is the existence of a natural two-sided market that does not require an opt-in negotiated counterparty. On the differential, the working-class household is naturally LONG (a widening differential corresponds to a relative loss for the household, which the policy-payoff compensates), while capital-holding households are naturally SHORT through the asset positions they already hold (their existing portfolios deliver the $R_a(t)$ leg, so their natural exposure is to a narrowing differential). The differential formulation therefore admits the same counterparty-matching structure that [Carter et al. \(2017\)](#) characterise for index-insurance arrangements: the index payoff clears the counterparty-availability problem that obstructs single-aggregate constructions, because the counterparty is not negotiated into a new position but is structurally exposed to the index already. The counterparty-matching argument is conjectural at the present stage; we flag the formal calibration of the differential's joint-leg correlation structure and FWER posture as the methodological question in [Section 7.3](#).

§3.3 Calibration of the continuous-time differential

The differential $Y_{\text{inequality}}(t)$ is treated as a continuous-time stochastic process. Its calibration — choice of the component variables that constitute each leg, weighting within each leg, joint-leg correlation structure, and treatment of regime switches — is deferred to [Section 7.3](#) as an explicit ask for the supervisor's input. The project's current iteration roster has not yet attempted full joint calibration; closed and active iterations estimate the response of single Y variables to single X lags as a precursor exercise. The parametric-cost-insurance reading developed above does not change the mathematical object being calibrated; it changes the standard against which the calibration's adequacy is judged, and that standard is the underwriting-discipline standard developed across [Carter et al. \(2017\)](#), [Mahul and Stutley \(2010\)](#), and [Clarke \(2016\)](#).

§4. On-Chain Settlement Substrate: Primer for the Non-Specialist Reader

This section introduces the on-chain settlement venue at the level required for the supervisor to evaluate the empirical record presented in [Sections 5 and 6](#). We anchor each on-chain construct to its closest traditional-finance analogue at first introduction and avoid implementation detail throughout. We emphasise at the outset that the empirical $\hat{\beta}$ -validation work the supervisor

is reviewing is conducted on off-chain data series (DANE, Banrep, GEIH); on-chain content is contextual, not evidentiary.

§4.1 What “On-Chain” Means

In the present document, an instrument is said to settle *on-chain* when the rule-set governing its cash flows — the conditions under which obligations are recognised, netted, and discharged — is recorded on a public ledger and executed by a programmable settlement engine. The closest traditional-finance analogue is a *clearinghouse with a public, verifiable, programmable rule-set*: the contractual specification that a clearinghouse keeps in its internal rulebook is, in this setting, a publicly inspectable object whose execution is automatic and observable to all market participants.

Two properties of this venue are load-bearing for what follows. First, settlement is *deterministic* given the rule-set and the observable inputs; there is no discretionary intermediary that must be coordinated with at trade time. Second, the rule-set is *programmable in advance*: payoff geometries that would conventionally require a bespoke ISDA confirmation can be encoded once and reused at scale. We emphasise that nothing in this section depends on the implementation language of the rule-set; the reader is asked only to accept that a programmable, publicly-auditable clearinghouse exists as the settlement backdrop for the instrument family the framework targets.

§4.2 Panoptic: Perpetual Options on Liquidity-Provider Positions

Panoptic is a deployed protocol that issues perpetual option-like instruments written against concentrated liquidity-provider (LP) positions on the Uniswap v3/v4 automated market makers (Lambert et al., 2023; Panoptic Labs, 2026; Adams et al., 2021, 2024). The traditional-finance analogue is a *perpetual-future-on-options instrument* in which the option strike is encoded as the price range over which a market-making position is concentrated, and the option premium accrues continuously as a streaming funding payment rather than as a one-shot premium at trade inception.

The economic content can be stated without recourse to implementation. A concentrated LP position, in the Uniswap v3/v4 sense, is a two-asset market-making commitment over a bounded price interval $[P_\ell, P_u]$. Inside the interval, the position behaves as a passive market maker; outside, it converts entirely to one of the two assets. The payoff of such a position, viewed as a function of the terminal price P_T , is a piecewise-concave curve — a long-gamma exposure inside the interval and a directional position outside it. Panoptic’s contribution is to allow market participants to take *either side* of this payoff continuously: a short position in the LP curve (sold to a long counterparty for a streaming premium) delivers a payoff geometry equivalent to a long perpetual option, with the strike, range, and convexity controlled by the chosen $[P_\ell, P_u]$ bracket and the asset pair. Standard option structures — covered calls, range LPs, perpetual puts, and straddles — are all expressible as combinations of such positions.

For the framework’s purposes, three properties matter. First, the instruments are *convex* in the underlying price: convexity is the load-bearing property required for the (Y, M, X) instrument

family articulated in Section 2 and the inequality-differential lemma of Section 3. Second, the instruments are *perpetual*: they do not expire on a fixed calendar date and therefore admit indefinite roll-without-rebooking, which is operationally relevant for a wage earner paying a small recurring premium out of wage income. Third, the instruments are *permissionless*: any market participant who can interact with the public ledger can take either side, without bilateral counterparty negotiation. We do not litigate liquidity adequacy at the venue here; Section 4.5 returns to the question of how the framework treats the gap between empirical $\hat{\beta}$ -validation and live deployment liquidity.

§4.3 Mento Purchasing-Parity Stablecoins

Mento is a deployed reserve-backed stablecoin protocol on the Celo public ledger (Mento Labs, 2026a,b). As of the manifest dated 2026-04-27, the Mento V3 deployment maintains a roster of twelve country-currency-pegged claims, of which the framework treats five as load-bearing for the target populations under iteration: USDM (United States dollar), EURM (Euro), BRLM (Brazilian real), KESM (Kenyan shilling), and COPM (Colombian peso). Each claim is, for the supervisor's purposes, a *digital country-currency-pegged liability of the protocol*, with the peg maintained by an on-chain reserve mechanism. The closest traditional-finance analogue is a narrow-purpose digital currency board: the protocol holds a portfolio of reserve assets and exchanges fixed amounts of the pegged claim against a basket of redemption assets at the protocol-determined reference rate, mediated by a routing contract referred to in the manifest as the Mento Broker.

The empirical implication for the framework is decisive and is the principal reason Mento appears in this document at all. Consider a working-class Colombian household contemplating a hedge against a macro risk X that threatens its purchasing power. A hedge denominated in United States dollars settles in a currency the household does not consume; the household then bears residual COP/USD exchange-rate risk between hedge settlement and consumption. A hedge denominated in COPM settles directly in a claim pegged to the household's consumption currency, eliminating that residual exchange-rate exposure by construction. The hedge is, in this sense, *purchasing-parity preserving* at the unit-of-account level. The same construction applies to BRLM for Brazilian populations, KESM for Kenyan populations, and so on across the Mento basket.

We note one item of housekeeping that may bear on supervisor questions about on-chain attribution. There exist two distinct Colombian-peso-named claims on the Celo ledger: the Mento-protocol-native COPM (canonical address `0x8A567e2a...4041eA`) and a separate fintech-issued COPM-MINTEO claim (address `0xc92e8fc2...1d5606`) that is not Mento-protocol-native and is therefore out of scope for any framework-level claim about Mento user behaviour. The audit-trail discipline of the framework treats these as non-substitutable.

§4.4 Political-Receptivity Context

In late January 2025, a tariff standoff between the United States executive and the Colombian executive produced a documented forty-eight-hour spike in new account formation at Litio, a Colombian fintech provider of USD-pegged stablecoin accounts; the spike was on the order of one hundred percent above prior baseline (per the project's internal reference document

OFFCHAIN_COP_BEHAVIOR.md, which the same event-window has been used as a structural-break event-dummy across multiple iteration specifications). This is a *user-side* responsiveness signal: Colombian retail moved into a USD-stablecoin rail during a political dollar-access shock. We include the observation as anecdotal context for the framework’s implicit hypothesis that working-class populations facing macro-risk shocks display measurable willingness to engage on-chain instruments when the off-chain banking rail becomes congested or politically encumbered. We emphasise that this observation is *not* a presidential transaction; earlier draft language asserting that the Colombian president had publicly transacted on-chain was unsourced and has been removed. The receptivity signal is anecdotal frame only; it is not load-bearing for any empirical or methodological claim elsewhere in this document.

§4.5 Critical Disclaimer: Empirical Validation is Off-Chain

The framework’s empirical $\hat{\beta}$ -validation work — the verdicts recorded in Section 6 and the methodology of Section 5 — is conducted on off-chain data series. The Pair D PASS verdict, the dev-AI Stage-1 in-progress estimation, the FX-vol-CPI-surprise FAIL, the Phase-A.0 EXIT, and the P1 PARKED record all rest on data assembled from the Departamento Administrativo Nacional de Estadística (DANE), the Banco de la República de Colombia (Banrep), and the Gran Encuesta Integrada de Hogares (GEIH). On-chain data series enter the framework only at later stages of candidate X_d signal-design, and their role there is auxiliary and re-pivoting in light of recent verification work (see Section 6, dev-AI Stage-1 and Phase-A.0 EXIT entries).

The settlement venue — Panoptic on a Mento-denominated pair — is the *deployment target* once a (Y, X) pair passes Stage-1 empirical validation. Stage-1 validation does not require any on-chain interaction; Stage-2 instrument design (M -sketch) does not require live liquidity; only Stage-3 deployment requires real liquidity-provider capital. The supervisor’s review concerns the econometrics — the Stage-1 layer. Sections 5, 6, 7, and 8, together with Appendices C and D, are readable end-to-end without further on-chain familiarity.

§4.6 Reader Takeaways

1. On-chain perpetual options exist and are tradable today, via the Panoptic protocol written against Uniswap v3/v4 LP positions (Panoptic Labs, 2026).
2. Purchasing-parity stablecoin denominations exist, via the Mento protocol’s roster of country-currency-pegged claims (USDM, EURM, BRLM, KESM, COPM) (Mento Labs, 2026b).
3. The empirical work the supervisor is reviewing is off-chain econometrics on DANE, Banrep, and GEIH series.
4. The remainder of this document does not require on-chain familiarity; the reader may proceed directly to Section 5.

§5. Empirical Methodology

This section describes the estimation discipline that has governed every iteration in the Abrigo log. The framework treats Stage 1 (empirical risk validation) as a stand-alone testing exercise: an iteration either produces a positive, statistically detectable composite coefficient on a pre-pinned lag panel of the macro risk X for the pre-pinned outcome Y , or it does not. The settlement geometry M on which an iteration would deploy is out of scope for the verdict. The methodology below therefore reads as a small, fixed kit of econometric primitives applied identically across iterations, with a strict pre-registration discipline that fixes every nontrivial design choice before the data are touched.

§5.1 Pre-registration discipline

For every iteration we author a design specification that pins, in textual form and *before any data ingest*, the following objects: the operational definition of Y ; the construction of X ; the lag set; the functional form of the dependent variable; the primary inference procedure; the robustness arms; the verdict thresholds; and the falsification tree. The specification is hashed (sha256) at commit time and the hash is propagated through the implementation plan, the result notebooks, and the verdict artifact. Any post-data revision recomputes the hash and triggers a two-wave independent review (Reality Checker plus a purpose-matched specialist) before further execution. Within an iteration, threshold tuning post-hoc is treated as silent fishing and is forbidden. This discipline is in the spirit of the pre-analysis-plan literature (Olken, 2015a; Christensen and Miguel, 2018b) adapted to an observational-econometrics setting in which a public registry analogue is not yet adopted by the project; whether to register against an external registry is one of the open questions we address to the supervisor in Section 7.

§5.2 Anti-fishing invariants

Three numerical invariants currently bind every iteration:

- Minimum sample size after window compliance and lag-window loss: $N_{\min} = 75$ observations.
- Minimum power at the pre-registered minimum detectable effect size: $\text{Power}_{\min} = 0.80$.
- Minimum detectable effect size: $\text{MDES} = 0.40$ standard deviations of Y .

These three numbers were inherited from the Phase-A.0 specification chain (Rev-5.3.1) and have *not* been re-derived per iteration's actual N , residual autocorrelation structure, or HAC-adjusted effective sample size. We flag this inheritance explicitly here, present whatever back-of-the-envelope derivation is feasible at writing time in Appendix D, and address the defensibility of the three numbers to the supervisor as a dedicated open question in Section 7 (§7.4). The document is asking the supervisor whether these thresholds are defensible against the empirical record of the five closed iterations, not asserting that they have been validated.

§5.3 Functional form and primary inference

For each iteration we test a one-sided null on the composite coefficient across a pre-pinned lag panel:

$$H_0 : \beta_{\text{composite}} \leq 0 \quad \text{vs.} \quad H_1 : \beta_{\text{composite}} > 0, \quad (1)$$

with the sign expectation grounded in the literature anchor of the iteration’s transmission chain (never inferred from the data). For the two monthly-cadence iterations that share the COP/USD lag panel (Pair D and dev-AI Stage-1), the primary specification is:

$$Y_t^{\text{logit}} = \alpha + \beta_6 \log(\text{COP_USD}_{t-6}) + \beta_9 \log(\text{COP_USD}_{t-9}) + \beta_{12} \log(\text{COP_USD}_{t-12}) + \varepsilon_t, \quad (2)$$

where $Y_t^{\text{logit}} = \log(Y_t/(1 - Y_t))$ is the logit of an interior-bounded employment share. The composite of interest is $\beta_{\text{composite}} \equiv \beta_6 + \beta_9 + \beta_{12}$; its hypothesis is tested via the linear-restriction variance $\widehat{\text{Var}}(c'\hat{\beta}) = c'\hat{\Sigma}c$ with $c = (0, 1, 1, 1)'$, equivalent to the Wald-restriction variance and not a delta-method object. Lags 6, 9, and 12 of the log-FX series are highly serially correlated at monthly frequency; individual coefficients have wide intervals, but the composite is precise because the off-diagonal entries of $\hat{\Sigma}$ are negative under positive collinearity, which deflates $\widehat{\text{Var}}(c'\hat{\beta})$ below the sum-of-individual-variances intuition. A reading that “individual lags weren’t significant” is methodologically incorrect: the test is on the composite alone.

The logit transform is justified on the standard textbook (Hansen, 2022a, §16.4) grounds for an interior-bounded share whose empirical range lies away from 0 and 1. The fractional-response (Papke and Wooldridge, 1996) and beta-regression (Ferrari and Cribari-Neto, 2004) alternatives are not pre-committed because the iterated Y ranges (Pair D [0.55, 0.75]; dev-AI Stage-1 [0.014, 0.031] realized) are well-interior to $(0, 1)$; a raw-share specification is preserved as a robustness arm rather than as the primary.

Primary inference convention. The primary standard error is the homoskedastic OLS standard error per the project’s anti-overclaim discipline: HAC bandwidth selection introduces a post-data degree of freedom that the pre-registration regime rules out from the primary tier. HAC inference is run as the pre-pinned R4 robustness arm (see §5.4) at a fixed lag-truncation chosen at spec-authoring time. Where the verdict label is invariant under the OLS and HAC standard errors, headline numbers report the *more conservative* of the two SE conventions; where the two would diverge across the verdict boundary, the OLS-primary SE governs the verdict and the HAC SE is reported as the R4 row. In the closed Pair D iteration the HAC SE produced a marginally smaller p-value than the OLS SE, so the headline reports the larger HAC SE as the more-conservative anti-fishing convention without re-routing the verdict; in the closed dev-AI Stage-1 iteration, primary OLS and R4 HAC standard errors agree on the FAIL verdict label and the headline reports the HAC SE on the same more-conservative-of-the-two convention. The verdict label has been invariant across the two SE conventions in every iteration that has closed to date.

§5.4 Robustness inference (HAC)

The robustness HAC standard error is the Newey-West heteroskedasticity- and-autocorrelation-consistent estimator (Newey and West, 1987) with *fixed lag truncation L chosen at spec-authoring time per iteration*, not a data-driven automatic rule. The bandwidth values that have been used to date are:

- Monthly cadence (Pair D, dev-AI Stage-1): $L = 12$. The truncation matches the longest right-hand-side lag in the spec — the twelve-month offshoring contracting cycle that motivates the lag panel.
- Weekly cadence (FX-vol-CPI primary): $L = 4$. The truncation was set at spec-authoring time on the basis of the in-sample residual autocorrelation diagnostic that closed at $L = 4$ in the Phase-A.0 pipeline; a robustness sensitivity at $L = 12$ is pre-registered as a fat-tail diagnostic, not as the primary.

Automatic-bandwidth selection rules (Newey and West, 1994; Andrews, 1991) are explicitly not used: a data-driven bandwidth introduces a post-data degree of freedom that the pre-registration discipline rules out.

§5.5 Robustness arms

For monthly iterations (Pair D and dev-AI Stage-1) the spec text pre-pins four single-row alternatives, each varying exactly one design choice from the primary; multi-dimensional re-specification (varying two or more dimensions simultaneously) is forbidden:

- R1 – regime-dummy alternative to the methodology-break treatment (e.g., 2021 GEIH *empalme* dummy in place of the DANE splice factor).
- R2 – sector-cut narrowing alternative on the dependent variable (Pair D: Sections J + M + N as a BPO-narrow cut; dev-AI Stage-1: Section M as a sibling-narrow comparator to Section J).
- R3 – functional-form alternative (raw share in place of logit).
- R4 – HAC inference per §5.4, substituted for OLS standard errors.

The closed FX-vol-CPI iteration uses a wider thirteen-row pre-registered sensitivity universe, of which the A1 monthly-cadence and A4 release-day-excluded rows are two; the universe is pinned via a `sensitivity_preregistration_hash` on the panel fingerprint artifact. Free lag tuning is forbidden in all iterations.

A coarse stability classifier maps the four-row vector to one of three labels: AGREE (zero R-rows produce a sign-flipped composite relative to the primary), MIXED (one or two flips), or DIS-AGREE (three or four flips). DISAGREE triggers a special verdict label `SUBSTRATE_TOO_NOISY`, regardless of the primary’s significance, and routes the iteration to a methodology-revision branch rather than to verdict assignment. We additionally note that the R4 HAC arm is a serial-correlation hedge and not, in itself, a heteroskedasticity hedge: the OLS-primary inference as-

sumes conditional homoskedasticity, and whether to add a White / sandwich heteroskedasticity-only diagnostic as a fifth pre-pinned arm is one of the methodology questions we put to the supervisor in Section 7 (§7.4).

§5.6 Persistent-regressor posture and Stambaugh bias

The right-hand-side regressor in the monthly-cadence iterations is $\log(\text{COP_USD}_{t-k})$, which is empirically near-unit-root at the monthly frequency on the realized 2014–2026 sample window. The project’s identification posture is predictive-regression in the strict Granger-predictive sense (see §5.9); the closed FX-vol-CPI iteration’s pre-registered T1 exogeneity test rejected at $F = 15.12$, $p \approx 1.3 \times 10^{-9}$ and consequently the project does not interpret $\hat{\beta}$ as an impulse-response.

The predictive-regression posture, however, is known to suffer small-sample bias when the right-hand-side regressor is persistent and the residual is correlated with the regressor’s innovation (Stambaugh, 1999); the bias does not vanish in finite samples even under correct specification, and recent econometric work (Phillips, 2014) has extended the asymptotic theory to near-unit-root and mildly explosive regressors. The Pair D and dev-AI Stage-1 iterations realized $N = 134$ monthly observations, which is modest-but-not-tiny on the conventional Stambaugh-bias scale; the bias-correction literature suggests the order- $1/N$ bias is small relative to the realized HAC standard error in both iterations, but the project has *not* formally implemented a Stambaugh-corrected estimator at writing time. This is one of the leading methodology open-questions we put to the supervisor in Section 7 (§7.1, identification ask, and §7.4, invariants ask): whether the predictive-regression posture is sufficient as Stage-1 verdict machinery, or whether Stambaugh-correction (or equivalent near-unit-root-aware inference) should be elevated from a deferred methodology improvement to a pre-pinned R-arm before the next iteration spec-pin.

§5.7 Verdict mapping

The mapping is deterministic. Given the primary tuple $(\hat{\beta}_{\text{composite}}, \hat{p}_{\text{one-sided}}, R\text{-consistency})$ and the residual-moment tuple $(\hat{s}_3, \hat{s}_4)$, every leaf returns exactly one verdict:

1. If the realized sample size N falls below $N_{\min} = 75$, the iteration HALTs (see §5.8).
2. If R-consistency is DISAGREE, the verdict is SUBSTRATE_TOO_NOISY.
3. Otherwise:
 - $\hat{\beta}_{\text{composite}} > 0$ and $\hat{p} \leq 0.05$ returns PASS.
 - $\hat{\beta}_{\text{composite}} > 0$ and $\hat{p} \in (0.05, 0.20]$ returns ESCALATE Clause A.
 - Clause B fires when $|\hat{\beta}_{\text{composite}}/\widehat{\text{SE}}| < 0.5$ and either $|\text{skew}(\hat{\varepsilon})| > 1.0$ or *Fisher excess kurtosis* $\hat{s}_4 \equiv \hat{\kappa}_4 - 3 > 3.0$ (i.e., raw fourth standardized moment $\hat{\kappa}_4 > 6$; a normal residual has Fisher excess kurtosis 0 and the threshold flags substantially leptokurtic tails). When Clause B fires the verdict is ESCALATE Clause B; otherwise FAIL.
 - $\hat{\beta}_{\text{composite}} \leq 0$ and $\hat{p} \leq 0.05$ returns FAIL with no Clause B evaluation: a wrong-signed significant estimate falsifies H_1 , and convex-payoff escalation, which is grounded in

the positive sign expectation, does not apply.

The numerical thresholds 0.5, 1.0, and 3.0 in Clause B are pinned at spec-authoring time and are not adjustable post-data; the 3.0 threshold is on the *Fisher* (excess-over-normal) convention, not on the Pearson raw-fourth-moment convention. The ESCALATE branches trigger a pre-authorized convex-payoff suite (a $\tau = 0.90$ quantile regression on the central lag, a GARCH-X mean-equation regression, and a peaks-over-threshold extreme-value regression on the upper-tail residuals); each suite leg has its own one-sided p-value at $\alpha = 0.10$ and the disjunction over the three legs determines ESCALATE-PASS or ESCALATE-FAIL. The disjunction structure is defended against multiple-testing critique on the grounds that each leg estimates a distinct distributional-moment parameter that maps to a distinct convex-instrument design, rather than three restatements of the same parameter.

§5.8 HALT and disposition discipline

When the spec text contradicts the data — examples already encountered include the DANE RAMA4D_R4 schema-stability HALT in the Pair D ingest and the realized cell-size and share-magnitude gap in the dev-AI Stage-1 ingest — the iteration HALTs by raising a typed exception. A disposition memo is filed enumerating at least three pivot options; the user adjudicates; a CORRECTIONS block lands in the next specification revision documenting the disposition, the chosen pivot, and the rationale; and a three-way independent review is run before further execution. Auto-pivot is forbidden, and post-hoc threshold tuning is forbidden. The discipline is documented in the project’s internal feedback memory under the *pathological halt anti-fishing checkpoint* rule.

§5.9 Identification posture

Every estimate produced by the framework to date is correlation-identified under predictive-regression posture. Wherever an exogeneity test has been run as a diagnostic — the FX-vol-CPI iteration ran a T1 exogeneity test that rejected at $F = 15.12$ and $p \approx 1.3 \times 10^{-9}$ — the result has reinforced the predictive-regression reading: $\hat{\beta}$ describes how past information predicts current realized volatility, not how an exogenous surprise causes future realized volatility. The distinction is the standard Granger-predictive versus structural-causal distinction; the document does not use $\hat{\beta}$ as an impulse-response. Whether to move to instrumental-variable, event-study, or structural VAR identification before declaring a Stage-1 PASS sufficient for Stage-2 dispatch — and whether the persistent-regressor posture noted in §5.6 requires Stambaugh-bias-corrected inference at the primary tier — is the leading open question we put to the supervisor in Section 7.

§6. The Abrigo Log: Record of Closed and Active Iterations

This section records the five iterations the project has run to date. Each iteration is presented in the same nine-field schema — target population; (Y, X) pair with operational definition; sign expectation and literature anchor; primary specification; sample; verdict; headline numbers; key caveats from independent review; and pre-registration sha256 chain. The section closes with a

load-bearing caveat on cross-iteration multiple-comparison structure that we develop further as a supervisor question in §7 (§7.5).

§6.1 Pair D — BPO offshoring through COP/USD lag

Target population. Colombian young workers aged 14–28 (statutory youth band per Ley 1622 de 2013) employed in the broad services aggregate.

(Y, X) pair. Y_t is the share of young workers (14–28) whose primary occupation falls in CIIU Rev. 4 A.C. Sections G–T, computed monthly on the DANE GEIH micro-data with `FEX_C_2018` expansion factors and the DANE *empalme* factor applied to the 2015–2020 segment for Marco-2005-to-Marco-2018 reconciliation; the dependent variable is the logit of the raw share. X_{t-k} is $\log(\text{COP_USD}_{t-k})$ on the Banrep TRM end-of-month spot, for $k \in \{6, 9, 12\}$ months.

Sign expectation and literature anchor. Positive. The transmission chain is Baumol cost disease (Baumol and Bowen, 1966; Baumol, 2006) composed with US-Colombia BPO wage arbitrage (Mendieta-Muñoz, 2017b) and the standard six-to-twelve-month offshoring contracting cycle documented in the Philippines comparator (Beerepoot and Hendriks, 2013; Errighi et al., 2017); displaced labor reallocates into services on the Rodrik-McMillan premature-deindustrialization channel (Rodrik, 2016a; McMillan and Rodrik, 2011a).

Primary specification. Equation (2) with composite $\beta_{\text{composite}} = \beta_6 + \beta_9 + \beta_{12}$; one-sided test at $\alpha = 0.05$; OLS-homoskedastic inference primary, HAC($L = 12$) inference as robustness arm R4.

Sample. $N = 134$ monthly observations, window 2015-01-31 through 2026-02-28. The window’s lower bound is the result of two CORRECTIONS-block revisions documented in the spec: an initial 2008-01 lower bound was shortened to 2010-01 after a schema-stability HALT on the DANE *empalme* nota-técnica coverage, and then to 2015-01 after a second HALT under the realization that the *empalme* files for the 2010–2014 segment ship CIIU Rev. 3 codes in the `RAMA4D` column despite the column header naming. The 2015-01 lower bound is the DANE-canonical Rev. 4 value-content-verified window. The $N = 134$ realized sample is well above the $N_{\min} = 75$ floor.

Verdict. PASS, 2026-04-28.

Headline numbers. $\hat{\beta}_{\text{composite}} = +0.13670985$; HAC standard error 0.02465; $t = +5.5456$; one-sided $p \approx 1.46 \times 10^{-8}$. Robustness rows R1–R4 produced zero sign flips relative to the primary; the four-row classifier returned AGREE.

Key caveats from independent review. Five flags from the Reality Checker review of the closed iteration are inherited as load-bearing constraints on any downstream framing:

1. The estimate hedges the *correlation* between lagged COP/USD and the broad-services young-worker share. It does not identify the BPO transmission channel as the operative mechanism: any narrative causal interpretation of the BPO chain is conjectural and is not supported by the Stage-1 design.
2. The lag effect is concentrated at the six-month horizon ($\hat{\beta}_6 \approx +0.109$, approximately 80% of the composite) rather than uniformly spread across the 6–12-month contracting window. The narrative that the project carries forward is “concentrated at the six-month horizon, within the six-to-twelve-month window,” not “spread uniformly across the window.”
3. The window-narrowing path through the two CORRECTIONS blocks over-represents three macroeconomically distinctive regimes (post-2014 oil-price collapse, the COVID shock, and the 2022–2024 Federal-Reserve tightening cycle); the regime-mix concern is a residual identification concern beyond the four pre-registered robustness arms.
4. The verdict is sensitive to a single off-spec design choice. An orchestrator-brief variant that included a `marco2018_dummy` in the primary — not in the spec text, where the dummy is the R1 arm only — produces $\hat{\beta}_{\text{composite}} = +0.0815$ at one-sided $p \approx 0.080$, which would map to ESCALATE Clause A under the verdict tree of §5.7. The spec text ran as authoritative because the spec sha256 was pinned 9.5 hours before Phase-2 dispatch; the off-spec variant is preserved in the artifact `primary_ols.json` under the field `off_spec_sensitivity_orchestrator_brief` for audit. The document carries this verdict-sensitivity record forward as part of the framework’s decision log.
5. *Compositional reframe surfaced by the dev-AI Stage-1 closed iteration (2026-05-06)*. The Pair D PASS was estimated on the broad CIU Sections G–T cut and was, at spec-pin time, interpreted under a “broad-services hedging the BPO transmission chain” reading consistent with either a Section-J ICT-narrow mechanism or a Section-M-style consultancy / scientific R&D / administrative-services mechanism aggregated up. The dev-AI Stage-1 iteration’s Section-J primary closed $\hat{\beta}_{\text{composite}} = -0.14613$ (HAC($L = 12$) SE 0.0847; $t = -1.726$; one-sided $p \approx 0.958$ on the pre-registered $H_1 : \beta > 0$) — a sign-flipped FAIL — and its R2 sensitivity arm on Section M closed $\hat{\beta}_{\text{composite}} = +0.45482801$ ($t = +4.73$; $p = 1.13 \times 10^{-6}$) at the same monthly cadence on the same lag panel. Read jointly with Pair D, this pattern strongly suggests Pair D’s PASS was carried by the Section-M-style sub-aggregate (consultants, legal, accounting, scientific R&D, administrative services) rather than by Section-J ICT-narrow tech offshoring. The spec §9.16 compositional- accounting concern is therefore *empirically resolved as flagged-not-resolved*: a formal (Sections G–T \ J) decomposition pre-registered as an R5 robustness row remains deferred for a future iteration, and any Stage-2 dispatch off the Pair D PASS must inherit this compositional caveat. Pair D’s headline numbers stand verbatim under the original spec; the interpretation of the operative sub-aggregate is what has sharpened.

Pre-registration sha256 chain. v1.1 → v1.2 → v1.2.1 → v1.3 → v1.3.1 → governing hash 964c62cca0...ef659. The full hash chain and revision history are reproduced in Appendix C.

§6.2 dev-AI Stage-1 — Section J ICT through COP/USD lag

Target population. Colombian young workers (14–28) employed in CIHU Rev. 4 A.C. Section J (Information and Communication, Divisions 58–63: publishing, motion picture and video, programming and broadcasting, telecommunications, computer programming and consultancy, information service activities). The population is a strict subset of the Pair D Sections G–T cut.

(Y, X) pair. Y_t is the Section-J young-worker share computed on DANE GEIH micro-data with the same monthly cadence, `FEX_C_2018` expansion factors, and `RAMA4D_R4` sector coding as Pair D, restricted to Section J via the published DANE Section table; the dependent variable is the logit. X_{t-k} is the same Banrep TRM end-of-month-spot panel as in Pair D, with the same lag set $k \in \{6, 9, 12\}$; the panel back-extends X to 2014-01 to preserve Y starting at 2015-01.

Sign expectation and literature anchor. Positive. The mechanism was the same Baumol-arbitrage-offshoring chain as Pair D narrowed from BPO-flavored services-broad to ICT-narrow tech-services offshoring; the same canonical references applied (Mendieta-Muñoz, 2017b; Beerepoot and Hendriks, 2013; Errighi et al., 2017; Rodrik, 2016a). As recorded below, the sign expectation was *rejected at conventional significance* by the realized estimate.

Primary specification. Equation (2) restated with the Section-J Y ; one-sided test at $\alpha = 0.05$; OLS-homoskedastic inference primary, HAC($L = 12$) inference as robustness arm R4.

Sample. $N = 134$ monthly observations, window 2015-01-31 through 2026-03-31, well above the $N_{\min} = 75$ floor.

Verdict. CLOSED FAIL, 2026-05-06.

Headline numbers. $\hat{\beta}_{\text{composite}} = -0.14613$ (HAC($L = 12$) standard error 0.0847); $t = -1.726$; one-sided $p \approx 0.958$ against $H_1 : \beta > 0$. The estimated sign is the opposite of the pre-registered expectation: the FAIL is a *sign-flipped* FAIL, not an underpowered failure-to-reject. The R-consistency classifier returned MIXED (three of four R-rows agree with the primary; only the R2 Section-M sensitivity sign-flips). The v1.0.2 pre-registered hardening (R1 and R3 must individually agree on sign for the verdict to clear PASS independently of the four-row aggregate) is consistent with the FAIL verdict at the negative sign. The pre-authorized escalation suite (§5.7) returned: D-i and D-ii FAIL on the positive-sign criterion; D-iii literally returned $\hat{\beta} = +0.113$, $p = 0.012$ but is *inadmissible* per the strict reading of §5.5 because the §3.3-trigger condition that would license escalation never fired. The user closed FAIL on 2026-05-06 per the strict §5.5 reading, with the D-iii literal positive preserved on the record as a Phase-3 artifact for a possible future Section-M iteration. (A spec-internal contradiction between §5.5 line 252 and §9.6 on the question of whether the escalation suite must run conditional on a primary-trigger or unconditionally is acknowledged on the project’s deferred-spec- reconciliation list as *spec v1.0.3 reconciliation flag*, to be resolved unambiguously before another §5.5 invocation.)

Key caveats from independent review.

1. *Cell-count and share-magnitude gap (CORRECTIONS- κ block).* Realized monthly cell counts for the Section-J young-worker stratum lay in [94, 267] with median 145; the ex-ante feasibility-memo baseline was [700, 1200]. Realized monthly raw shares lay in [0.014, 0.031] against an ex-ante range of [0.04, 0.10]. The logit derivative $d\logit(Y)/dY = 1/[Y(1 - Y)]$ at the realized share range maps to [33, 73], against a v1.0.1 across-support amplification ratio of 2.34; the spec was κ -tightened on this basis at v1.0.2 to require R1 and R3 individual sign agreement before any PASS verdict. The realized estimate cleared the κ -tightened reading at the *negative* sign — the substrate is not too noisy under the DISAGREE classifier (only one R-row sign-flips), so the verdict is routable rather than SUBSTRATE_TOO_NOISY.
2. *R2 Section-M sensitivity — compositional finding.* The R2 sensitivity arm narrowed Y from Section J to Section M (CIU Rev. 4 Sections 69–75: professional, scientific, technical, and administrative services) at the same lag panel and primary functional form, and produced $\hat{\beta}_{\text{composite}} = +0.45482801$ at $t = +4.73$, one-sided $p = 1.13 \times 10^{-6}$. This is a strong positive signal at a sister sub-aggregate to Section J on the same FX-lag panel. The framework reads this as a *candidate-next-iteration finding*, not as a Pair D rescue or a dev-AI Stage-1 rescue: the Section-M coefficient was not pre-registered as the primary at spec-pin time, and per the project’s anti-fishing discipline it cannot graduate to a PASS-equivalent claim from a robustness-arm posture. What it *does* resolve, jointly with the Pair D PASS and the dev-AI Stage-1 FAIL, is the spec §9.16 compositional-accounting ambiguity: Pair D’s broad-services PASS is *empirically consistent with* a Section-M-style consultancy-and-professional- services transmission chain rather than a Section-J ICT-narrow chain. A future iteration with Section M pre-registered as the primary, on a fresh spec-pin and the canonical four-row R-arm discipline, is the routed-forward path.
3. *Empalme residual bias surfaced.* The Phase-1 ingest diagnostic (Notebook 02 Trio 1 boundary_anomaly) returned TRUE: the logit- Y jump at the 2020-12 $\rightarrow$ 2021-01 boundary was +0.375, against an envelope of ± 0.335 (three times the historical month-to-month standard deviation). The R1 regime-dummy arm produced $\hat{\beta}_{R1} = +0.188$ at $t = +4.36$, reflecting an unexplained level shift in the post-2021 era. The DANE Marco-2018 empalme correction was applied to the primary as designed, but did *not* fully neutralize the methodology break at the Section-J narrow cut; the regime structure is informational diagnostic and the project explicitly does not use the $\hat{\beta}_{R1}$ sign as a primary-spec rescue lever. The empalme-residual phenomenon is an inherited methodology constraint for any future DANE-GEIH-based iteration.
4. *Per-user instrument design implication.* The Stage-1 rejection of the positive-sign expectation falsifies the transmission for which a per-user a_s instrument was the candidate Stage-2 deployment: the rejected sign means the dev-AI- paying population (Colombian young Section-J workers paying USD-denominated AI APIs) does not, at Stage-1, exhibit the FX-pass-through-to-employment pattern the convex hedge would settle. A SHORT-FX-style

instrument would be the opposite of what that population needs given USD-denominated AI tooling costs; consequently a per-user a_s construction on this transmission is not designable. The Section-M finding does not change this conclusion at Section-J granularity, and any Section-M follow-on iteration must pre-pin its own population definition before the Stage-2 instrument question reopens.

Pre-registration sha256 chain. $v1.0 \rightarrow v1.0.1 \rightarrow v1.0.2 \rightarrow$ governing hash `7c72292516...751f5a`.

§6.3 FX-vol-CPI surprise — Colombian CPI through TRM realised volatility

Target population. Colombian corporates and remittance recipients with FX-volatility exposure on the COP/USD pair around Colombian CPI publication.

(Y, X) pair. Y_t is the weekly realized volatility of the Banrep TRM end-of-day series, computed on the daily log-return panel and aggregated to the weekly cadence under a Friday anchor. X_t is the signed AR(1) CPI surprise: the monthly DANE CPI release minus a one-step-ahead AR(1) forecast on prior CPI prints, mapped to the weekly cadence by carrying the surprise across the release week and zeros otherwise.

Sign expectation and literature anchor. Positive on the classic CPI-announcement-volatility channel: a larger absolute surprise should generate larger TRM realized volatility around the release. The pre-registered primary tested the signed surprise rather than the absolute surprise, on the prior that the Colombian hyperinflation anchor produces an asymmetric response.

Primary specification. Weekly OLS of TRM realized volatility on the signed AR(1) CPI surprise plus a small set of macro-surprise controls; one-sided test at $\alpha = 0.05$; OLS-homoskedastic inference primary, HAC($L = 4$) inference under the gate-aggregation rule. The pre-registered sensitivity universe spans thirteen rows (A1–A12 plus seven cumulative cuts S1–S7) hashed via a `sensitivity_preregistration_hash` on the panel fingerprint artifact.

Sample. $N = 947$ weekly observations, window 2008-01-02 through 2026-02-23.

Verdict. CLOSED FAIL on the pre-registered weekly primary, 2026-04-19.

Headline numbers. $\hat{\beta}_{\text{CPI}} = -0.000685$ (HAC(4) standard error 0.001794); 90% HAC interval $[-0.003636, +0.002266]$ (contains zero); the gate criterion $\hat{\beta} - 1.28 \cdot \widehat{\text{SE}} < 0$ returns $-0.002981 < 0$, which is the FAIL trigger. Adjusted $R^2 = 0.193$ clears the pre-pinned floor; the macro-surprise regressors alone contribute approximately zero. Under the pre-committed weekly OLS specification, Colombian AR(1) CPI surprises do not produce a statistically detectable increase in COP/USD realized volatility across the sample window. We separately note that the pre-registered A1 monthly-cadence sensitivity returned $\hat{\beta} = +0.0152$ with a 90% interval $[+0.0057, +0.0246]$, and the A4 release-day-excluded sensitivity returned $\hat{\beta} = +0.0033$ with a 90% interval $[+0.0005, +0.0062]$. The A4 90% interval excludes zero uncorrected: the rejection at the one-sided $\alpha = 0.05$ convention is clean before any multiple-testing correction, even though

the lower bound (0.0005) is small in absolute magnitude; calling A4 “edge-of-significance” would be inaccurate. Neither A1 nor A4 is the iteration’s main finding: the pre-registered primary is FAIL and the pre-registered primary *is* the result. The two sensitivities are pre-registered robustness checks that returned positive significance at the 90% one-sided convention; they live in the iteration’s pivot record as candidate inputs to a future monthly-cadence-primary specification, not as a rescue of the weekly primary.

Key caveats from independent review. The pre-registered T1 exogeneity test rejected at $F = 15.12$, $p \approx 1.3 \times 10^{-9}$; the iteration is therefore a *predictive regression* in the strict sense of the Granger-predictive-versus-structural-causal distinction, and the estimated $\hat{\beta}$ describes how past information predicts current realized volatility, not how an exogenous surprise causes future realized volatility. We do not interpret $\hat{\beta}$ as an impulse-response. The HAC-versus-bootstrap reconciliation across the verdict tree returned AGREEMENT, ruling out a standard-error artifact.

Pre-registration sha256 chain. The iteration ran on the Phase-A.0 Rev 4 specification chain; the canonical artifacts are the panel fingerprint (`nb1_panel_fingerprint.json`) and the verdict file (`gate_verdict.json`), both governed by the `sensitivity_preregistration_hash` pinned in the panel fingerprint.

§6.4 Phase-A.0 remittance — Colombian remittance corridor through FX

Target population. Colombian household remittance recipients, indexed against the Colombian remittance corridor with FX pass-through.

(Y, X) pair. Y_t was the weekly TRM realized volatility panel (inherited from §6.3). X_t was a candidate on-chain remittance-flow proxy constructed from cCOP and COPm aggregate user activity on Celo, partitioned to attempt to isolate Colombian household-remittance flow from third-party-DEX activity, treasury roundtripping, and bot arbitrage.

Sign expectation and literature anchor. Positive on the remittance-volatility channel: large remittance-flow shocks should generate FX-vol shocks at the receiving corridor. The literature anchor was the parametric-insurance-counterparty-matching tradition.

Primary specification. Pre-registered weekly OLS with empirical kill-criteria k1 (intent: an X-driver fingerprint must be present in remittance event-window data) and k2 (data-source: the X proxy must measure Colombian household remittance, not a generic on-chain volume aggregate).

Sample. Window 2024-09 through 2026-04, weekly cadence (panel construction did not reach final N before the EXIT verdict fired).

Verdict. CLOSED EXIT_NON_REMITTANCE, 2026-04-24.

Headline numbers. The iteration exited at the kill-criteria step rather than producing point estimates. Task 11.F Axis-1 event-research returned zero of thirty peak-day flows that fingerprinted as remittance; approximately 87% of the surveyed flow decomposed into TRM arbitrage, treasury roundtripping, campaigns, and bot activity, with the residual a thin retail tail. Kill criterion k1 fired (intent non-remittance dominant); kill criterion k2 fired in part (the unfiltered Dune aggregate at query 7366593 did not measure Colombian household remittance).

Key caveats from independent review. The exit was load-bearing on the kill-criterion discipline. Without the pre-registered EXIT criterion, the rev-4 plan’s “non-stop filter iteration” policy would have run to a filter that appeared to “rescue” a remittance signal that was never captured upstream. The iteration carries forward the lesson that no post-hoc filter can reconstruct what was never measured upstream; any pivot must inherit the kill-criterion pattern.

A further caveat applies to subsequent on-chain attribution. A 2026-04-24 thesis proposed Carbon-DeFi basket-rebalancing volume as a candidate on-chain demand signal for an inequality-differential follow-on iteration; that attribution was retracted on 2026-04-27 after verification of the Mento V3 deployment manifest confirmed zero protocol-level integration between the Mento contracts and the Carbon contracts. Carbon `tokenstraded` events on Celo are pure third-party-DEX activity dominated by Bancor’s own arbitrage routers (383,303 distinct traders on the Mento Broker V2 swap event versus 147 on Carbon over comparable windows), not Mento Reserve user demand. The framework’s on-chain demand-signal candidate has been re-pivoted to native Mento Broker V2 swap events on the Broker contract `0x777A8255...4CaD`, and the supervisor-review document does *not* present Carbon-DeFi volume as the canonical Phase-A.0 finding. The Phase-A.0 EXIT verdict stands on the original (Y, X) pair as falsified-on-mechanism rather than failed-on-power.

Pre-registration sha256 chain. Phase-A.0 spec chain $\text{Rev 1} \rightarrow \text{Rev 1.1} \rightarrow \text{Rev 1.1.1}$ (all SUPERSEDED and RETIRED at EXIT) $\rightarrow \text{Rev 4.1}$ (kill-criterion-bearing); the on-chain attribution retraction is recorded in the framework memory at Rev-5.3.7.

§6.5 P1 Bittensor SN18 — event-study apparatus

Target population. The proprietary-AI policy-shock-exposed population (LATAM developers paying USD-denominated AI APIs is the downstream candidate; the apparatus was authored before the target-population narrowing).

(Y, X) pair. Y_t was a Bittensor SN18 (Cortex.t) alpha event-study indicator; X_t was a set of pre-pinned policy-event dates (Bittensor halving 2025-12-15; dTAO mainnet activation 2025-02-13; Cortex.t milestone events).

Sign expectation and literature anchor. Two-sided. The methodology baseline was the Maymin (2026, arXiv:2603.29751) event-study framework with explicit divergences pinned in the spec.

Primary specification. Event-study apparatus over a nine-cell verdict cube (verdict-eligibility crossed with robustness-consistency), Bonferroni-corrected at $\alpha_{\text{primary}} = 0.0167$ over the conjunctive family $P1 + P2 + P3$, with $N_{\text{min}}^{\text{events}} = 8$ and an asymmetric placebo gate.

Sample. Apparatus completed; data ingest deferred.

Verdict. PARKED, 2026-04-27.

Headline numbers. None: the apparatus is parked for the record; no graduating empirical claim was produced.

Key caveats from independent review. The slow-lane apparatus (three spec revisions, six verifier dispatches, a nine-cell verdict cube) was overkill for the project’s stage goal of identifying a single empirically validated risk and a positive- β confirmation before protocol design. The user pivoted to the simple- β fast-lane track (Pair D, dev-AI Stage-1) and parked P1 to preserve the intellectual capital of the spec work. Re-activation conditions are documented in the parking memory.

Pre-registration sha256 chain. Spec sha256 f855e036d3...b47aeab.

§6.6 Cross-iteration multiple-comparison caveat

The five iterations above are pre-registered individually but the *decisions* that produced them — which target populations to pursue, which (Y, X) pairs to spec-pin, which to drop after an EXIT, which to deepen — are themselves a meta-multiple-comparison structure. As of 2026-05-06 all five iterations are closed or parked (Pair D PASS; dev-AI Stage-1 FAIL; FX-vol-CPI FAIL; Phase-A.0 EXIT_NON_REMITTANCE; P1 SN18 PARKED). Pair D’s one-sided $p \approx 1.46 \times 10^{-8}$ is overwhelmingly significant against any reasonable [Benjamini and Hochberg \(1995\)](#) or [Romano and Wolf \(2005\)](#) correction applied across five iteration-level tests, but the FX-vol-CPI pre-registered sensitivities (A1 monthly with 90% interval $[+0.0057, +0.0246]$; A4 release-day-excluded with 90% interval $[+0.0005, +0.0062]$) and the dev-AI Stage-1 R2 Section-M finding ($\hat{\beta} = +0.455$, $p = 1.13 \times 10^{-6}$, but not pre-registered as a primary) are differentially robust to such a correction; the A4 lower bound at 0.0005 is small in absolute magnitude though the rejection is clean uncorrected. The Section-M finding sits as a candidate-next-iteration object rather than a graduated PASS, precisely because it was not pre-pinned as a primary at spec-author time. We therefore decline to make any aggregate claim about “what the project has established across five iterations.” What the project has established at the per-iteration level is recorded above, iteration by iteration, with the appropriate hedges. The framework’s posture toward iteration-selection bias and the appropriate familywise-error-rate or false-discovery-rate correction regime is the dedicated open question we put to the supervisor in Section 7 (§7.5).

§7. Open Methodological Questions for Supervisor Review

We present five methodological questions in priority order. The first four were enumerated at the brainstorming stage of the memorandum; the fifth was added on Wave-2 verification feedback to make the cross-iteration multiple-comparison concern explicit rather than allowing the supervisor to introduce it as an unanticipated objection. For each ask, we state the project’s current posture, the references that frame the question for an econometrician of the supervisor’s training, and the specific decision the supervisor’s response would unblock.

§7.1 Identification strategy

Current posture. The Stage-1 specification is OLS on lagged log-FX with HAC inference at fixed lag truncation L (see Section 5). The estimator $\hat{\beta}_{\text{composite}}$ is a predictive-regression coefficient; we do not treat it as an impulse-response. The PASS-verdict iteration (Pair D) flags identification non-uniqueness explicitly: the closure memorandum hedges the correlation between Section-G–T employment share and lagged COP/USD movement and does *not* assert a causal channel running through BPO contracting. The FAIL-verdict iteration (FX-vol-on-CPI-surprise) records a T1 exogeneity rejection at $F = 15.12$, $p \approx 1.3 \times 10^{-9}$, confirming that the FX-vol exercise is also predictive rather than structural.

Question for the supervisor. Should subsequent iterations move from a predictive-regression posture to a causally-identified one, and if so, what identification standard does the supervisor require before declaring a Stage-1 PASS sufficient for Stage-2 dispatch? Specifically:

- Does the supervisor require an instrumental-variable strategy? Candidate instruments under consideration include monetary-policy-rate decisions of the central bank issuing the consumption-basket currency, oil-price shocks (relevant for commodity-exporting populations), and US-side AI-API price changes (relevant for the dev-AI Stage-1 iteration). The relevant references are [Hansen \(2022b\)](#), Chapters 6 and 12, and [Wooldridge \(2010\)](#).
- Does the supervisor require an event-study formulation? The Phase-A.0 remittance work and the parked P1 Bittensor SN18 work both have event-study apparatus available; we have not promoted either to a Stage-1 primary in any closed iteration.
- Does the supervisor require a structural vector autoregression? The reference framework would be [Kilian and Lütkepohl \(2017\)](#); the project does not currently have a structural identification scheme that we believe defensible against that reference’s standards.

What the supervisor’s answer unblocks. A pinned identification standard for subsequent iterations, including dev-AI Stage-1’s eventual Stage-2 dispatch.

§7.2 Convex-payoff sufficiency

Current posture. Stage-1 estimates a mean slope. The internal working note `memory/project_abrigo_convex` records that mean- $\hat{\beta}$ identification is necessary but not sufficient for pricing payoffs that transmit

through the upper or lower tail of the conditional distribution rather than through the conditional mean. Under the parametric-cost-insurance reading developed in Sections 1 and 3, this insufficiency is the basis-risk question of [Clarke \(2016\)](#): a mean-calibrated index specification can leave the policy-holder materially exposed in precisely the tail states that motivate the demand for the policy in the first place.

Question for the supervisor (insurance-framing). The operative question is: *what empirical evidence does the supervisor require before this $\hat{\beta}$ -validation is sufficient to underwrite a parametric cost-insurance product on the validated X ?* We adopt the parametric-index-insurance literature ([Carter et al., 2017](#); [Mahul and Stutley, 2010](#); [Clarke, 2016](#)) as the methodological reference rather than the generic option-pricing literature, on the deliberate ground that the insurance literature imposes the stricter evidentiary standard. Where option-pricing methodology is concerned principally with the second moment of the underlying, the parametric-insurance methodology requires the underwriter to characterise basis risk against the policy-holder's realised loss ([Clarke, 2016](#)), the adverse-selection structure of the policy-holder pool ([Mahul and Stutley, 2010](#)), and the catastrophic-loss probability and reinsurance posture of the product ([Carter et al., 2017](#); [Mahul and Stutley, 2010](#)). Under that stricter standard, the specific Stage-1 questions become:

- Is variance-amplification evidence at the X -shock dates sufficient (for example, a GARCH- X specification with the validated X entering the conditional-variance equation), or is variance amplification merely necessary?
- Is full quantile-regression coverage required, with explicit estimation of upper- and lower-tail conditional quantiles of $Y \mid X_{t-k}$, so that the basis-risk envelope of [Clarke \(2016\)](#) is directly observable from the Stage-1 output?
- Is extreme-value-theory peaks-over-threshold tail-index estimation required, in the spirit of [Embrechts et al. \(1997\)](#), in order to bound the catastrophic-loss probability that [Carter et al. \(2017\)](#) and [Mahul and Stutley \(2010\)](#) treat as a first-order underwriting input?
- Should the dynamic conditional-correlation MGARCH specification of [Engle \(2002\)](#) be adopted as the standard joint-distribution model for the differential's two legs in Section 3, given that the joint-distribution behaviour determines the differential-leg basis risk?

The supervisor is invited to apply whichever subset of these standards he judges appropriate; the project's posture is to err on the side of the stricter insurance-underwriting reading rather than the more permissive option-pricing reading.

What the supervisor's answer unblocks. The Stage-1 exit criterion for Stage-2 dispatch under the parametric-cost-insurance product objective, and in particular the basis-risk evidentiary standard that Stage-1 output must meet before the validated X is treated as underwritable.

§7.3 Multi-Y differential calibration

Current posture. Lemma 1’s differential $Y_{\text{inequality}} = R_a - R_c$ requires constructing each leg as a panel of observable component variables. Closed and active iterations estimate single-Y, single-X responses; the project has not yet calibrated the joint-leg structure, and the within-iteration thresholds ($N_{\text{MIN}} = 75$, $\text{POWER}_{\text{MIN}} = 0.80$, $\text{MDES}_{\text{SD}} = 0.40$) were derived under a single-Y assumption.

Question for the supervisor. How should the joint-leg structure of $Y_{\text{inequality}}$ be calibrated? Specifically:

- Joint-leg correlation: what specification — a constant-correlation GARCH-type model, the dynamic conditional-correlation specification of Engle (2002), or a regime-switching extension — does the supervisor recommend for the conditional joint distribution of (R_a, R_c) ?
- Within-leg weighting: what method — fixed welfare-anchored weights (the working-class consumption basket), principal-components weights, or weights estimated from the panel itself — should determine the composition of each leg?
- Regime-switching: should the differential admit a Markov-switching mean-and-variance structure to accommodate the documented regime mix (post-2014 oil collapse, COVID, Fed-tightening) noted in the Pair D closure memorandum, or should regime indicators enter as exogenous covariates only?
- Familywise error: under the multi-Y panel implied by the differential, should familywise-error-rate control proceed under Bonferroni-Holm, or under the false-discovery-rate framework, given that within-leg components are expected to be highly pairwise correlated and Bonferroni is therefore likely conservative?

What the supervisor’s answer unblocks. The joint specification of the inequality-differential dynamics, which is a prerequisite for promoting Lemma 1 from a motivating proposition to a calibrated estimation target.

§7.4 Anti-fishing protocol defensibility

Current posture. The within-iteration anti-fishing invariants are $N_{\text{MIN}} = 75$, $\text{POWER}_{\text{MIN}} = 0.80$, and $\text{MDES}_{\text{SD}} = 0.40$ standard-deviation units of Y . These three thresholds were inherited from the Phase-A.0 Rev-5.3.1 specification and have *not* been re-derived under each subsequent iteration’s actual N and observed first-order autocorrelation structure of the residuals. Appendix D presents whatever power-derivation transparency is feasible at the time of writing; we flag the inheritance explicitly so that the supervisor’s review is on the actually-defensible numbers and not on the inherited ones.

Question for the supervisor. Are these three thresholds defensible against the empirical record of the five closed iterations, and what additional pre-registration discipline does the supervisor require? Specifically:

- Should the project register pre-analysis plans in a public registry, in the spirit of [Olken \(2015b\)](#) on pre-analysis plans in development economics and [Christensen and Miguel \(2018a\)](#) on transparency and reproducibility in economics?
- Should the project adopt the AEA RCT-registry analogue for observational pre-registration? The current discipline pins spec-text by sha256 hash inside the project repository; this is auditable internally but not externally registered.
- Should the three thresholds be re-derived per iteration, taking each iteration's observed N and HAC-corrected effective sample size as inputs to the power calculation, in the spirit of [Hansen \(2022b\)](#), § 6.18 on autocorrelation-corrected effective sample sizes?

What the supervisor's answer unblocks. The pre-registration infrastructure for subsequent iterations, including whether the inherited threshold triple should remain in force or be replaced.

§7.5 Cross-iteration FWER and iteration-selection bias

Current posture. The project has run five iterations and all five are closed or parked: one PASS (Pair D), two FAIL (one pre-registered-weekly-primary FAIL with two sensitivity-positive arms in FX-vol-on-CPI-surprise; one sign-flipped FAIL on the dev-AI Stage-1 narrow-ICT primary closed 2026-05-06), one EXIT (Phase-A.0 remittance), and one PARKED (P1 Bittensor SN18). No Stage-1 work is in progress as of the memorandum date. Each iteration's within-iteration pre-registration discipline is documented in Section 5; the *cross-iteration* structure is not pre-registered. Iteration-selection decisions — which (Y, X) pairs enter the roster, which are dropped after EXIT, which are deepened after PASS — are made at the project level and are not bound by an external pre-registration rule.

The numerical mass at the two tails across the five tests is asymmetric in a way that matters for any FWER-style correction. On the positive tail, the Pair D closure verdict's headline $p \approx 1.46 \times 10^{-8}$ is overwhelmingly significant against any reasonable Bonferroni correction across five iteration-level tests; the FX-vol-on-CPI-surprise sensitivity arms are far less robust to such a correction (the A1 monthly sensitivity reports a 90 % confidence interval of $[+0.0057, +0.0246]$; the A4 release-day-excluded sensitivity reports $[+0.0005, +0.0062]$, and the A4 lower bound at 0.0005 would not survive a meaningful familywise-error-rate adjustment). On the *negative* tail, the dev-AI Stage-1 primary delivered $\hat{\beta} = -0.146$ at $t = -1.726$, which we record as a high-information FAIL: the iteration was pre-registered with a positive-sign expectation, and the realisation is sign-flipped at the direction of the alternative rather than merely insignificant. A two-sided cross-iteration FWER procedure has different mass at the two tails of the iteration-result distribution, and the supervisor's preferred procedure should be specified to make the asymmetry explicit rather than letting a default Bonferroni argument flatten it.

Question for the supervisor. What familywise-error-rate or false-discovery-rate correction regime should be applied across the iteration history, given the directional asymmetry just described, and how should sensitivity-positive findings such as the FX-vol A1 and A4 results be

weighed under such a correction? In particular, should a sign-flipped FAIL at conventional significance be counted symmetrically with the PASS at the positive tail under the supervisor's preferred correction, or should the directional pre-registration on each iteration shift the mass allocation?

A second sub-question is methodologically prior. Iteration selection itself — the decision of which (Y, X) pair to specify, which to drop, which to deepen — is a multiple-testing structure that within-iteration pre-registration does not address. What mitigations does the supervisor recommend? Candidate mitigations under consideration include pre-registered iteration-budget caps committed at project initiation, an iteration-level alpha-spending function in the spirit of group sequential designs, and pre-registered iteration-allocation rules that do not depend on prior verdicts. We have no current basis for choosing among these and are explicitly asking for the supervisor's preferred form.

Sharpened sub-question — is the Pair D PASS one test, or one test conditional on a sub-aggregate selection? The dev-AI Stage-1 iteration's pre-specified Section-M sensitivity arm (professional/scientific/technical/admin services, CIU Rev. 4 Sections {69–75}) produced $\hat{\beta} = +0.455$ at $p \approx 1.13 \times 10^{-6}$, while the iteration's pre-registered Section-J narrow-ICT primary failed sign-flipped at $\hat{\beta} = -0.146$. The Section-M sensitivity was a sensitivity arm of the dev-AI iteration, not a pre-registered primary; we therefore do not present the Section-M finding as a free-standing positive verdict. We do, however, observe that the Section-M result is *consistent with* a compositional reading of the Pair D PASS under which the headline broad-services Section-G–T positive estimate reflected a Section-M-style transmission averaged across the broader G–T aggregate rather than a Section-J ICT-narrow signal. The internal specification **spec §9.16** flagged precisely this compositional ambiguity at Pair D's closure; the formal **§9.16(c)** (G–T minus J) decomposition is flagged-not-resolved per the dev-AI Stage-1 verdict memorandum, and the Section-M sensitivity arm is consistent with but not equivalent to that decomposition.

This sharpens the supervisor's FWER question rather than answering it. If the Pair D PASS is read as the resolution of one pre-registered test, the cross-iteration tally is one PASS out of five tests at the positive tail. If the Pair D PASS is read as one test conditional on a post-hoc-defensible-but-not-pre-registered choice of sub-aggregate within G–T (with the dev-AI Section-M sensitivity arm now suggesting that the sub-aggregate concentrating the signal is Section M rather than Section J), then the multiple-comparison ledger across the iteration history must additionally account for the implicit search across sub-aggregates of the broad-services panel. We flag this as a sharpened sub-question for the supervisor and do not litigate it here. The references for selection-on-observables and sample-selection correction in the spirit of [Heckman \(1979\)](#) are noted as one candidate frame for the supervisor's response, alongside the FWER and false-discovery-rate frames already cited in this section.

What the supervisor's answer unblocks. The project-level multiple-testing discipline that supplements within-iteration pre-registration; the appropriate weight to assign to the sensitivity-positive findings of the FX-vol FAIL-primary iteration; the appropriate weight to assign to the sign-flipped FAIL primary of the dev-AI Stage-1 iteration under a directionally-aware correc-

tion; and the appropriate weight to assign to the Pair D PASS once the sub-aggregate-selection question is taken into account.

§8. Request for Guidelines

The five questions of Section 7 are presented in our current priority ordering, but the ordering itself is one of the things on which we are seeking the supervisor’s guidance. We would value an explicit recommendation on the priority order in which subsequent iterations should address the five asks, and — where applicable — recommended numerical thresholds, identification standards, joint-distribution specifications, and cross-iteration multiple-comparison regimes.

We expect the supervisor’s response, once given, to translate into binding design constraints for subsequent iterations: the next iteration’s pre-registered specification will adopt the supervisor’s preferred identification strategy under Section 7.1; the Stage-2 exit criterion will adopt the supervisor’s preferred convex-payoff sufficiency test under Section 7.2; the joint-leg calibration of Lemma 1 will follow the supervisor’s preferred specification under Section 7.3; the within-iteration anti-fishing thresholds will be retained, revised, or augmented per the supervisor’s guidance under Section 7.4; and the cross-iteration multiple-comparison discipline will be parameterised per the supervisor’s recommendation under Section 7.5.

Framing pin. The implementation venue for the resulting products is on-chain settlement infrastructure, summarised in Section 4 (Panoptic perpetual options on Uniswap-V3/V4 liquidity-provider positions, denominated against Mento purchasing-parity stablecoins where appropriate); the *economic primitive*, however, is parametric cost insurance for an underserved population, not a derivative product offered to a general market. The supervisor’s review applies insurance-econometrics standards (Carter et al., 2017; Mahul and Stutley, 2010; Clarke, 2016) to the empirical work documented in Sections 5 and 6; the on-chain settlement layer is implementation infrastructure, not the object of review. The memorandum makes no implicit promises about either the form or the timing of the supervisor’s response; the commitments above are conditional commitments on our side, not anticipated commitments on the supervisor’s.

§A. Glossary

Econometric terms

HAC (Newey-West) Heteroskedasticity-and-autocorrelation-consistent standard error using the Hansen (2022b)-style Bartlett kernel at fixed lag truncation L ; used in the project as a robustness arm rather than as primary inference.

Composite- $\hat{\beta}$ Sum of lag-distributed slope coefficients across the pre-registered lag set, $\hat{\beta}_{\text{composite}} = \hat{\beta}_6 + \hat{\beta}_9 + \hat{\beta}_{12}$ for monthly iterations; the primary test statistic at Stage 1.

MDES Minimum detectable effect size; the smallest slope under which the pre-registered specification achieves the committed power at the committed significance level.

MDES_{SD} MDES expressed in standard-deviation units of Y ; pinned at 0.40 in the inherited Phase-A.0 Rev-5.3.1 invariant.

FWER Familywise error rate; the probability of at least one Type-I error across a family of tests.

FDR False-discovery-rate (Benjamini-Hochberg 1995); the expected proportion of false positives among the rejected hypotheses, used as the less conservative alternative to FWER under correlated tests.

Structural VAR Vector autoregression in which contemporaneous relationships among the endogenous variables are identified by economically-motivated restrictions; reference framework Kilian and Lütkepohl (2017).

Predictive regression Regression in which the right-hand-side variable carries past information about the left-hand-side variable but is not assumed exogenous in the structural sense; coefficients are predictive, not impulse-response.

Granger causality Statistical relation in which past values of one series help predict another conditional on the second's own lags; not equivalent to structural causation.

DCC-MGARCH Dynamic conditional-correlation multivariate GARCH specification of Engle (2002); candidate joint-distribution model for the two legs of the inequality differential.

EVT POT Extreme-value-theory peaks-over-threshold; tail-index estimation framework of Embrechts et al. (1997), candidate input to the Stage-1 convex-payoff sufficiency test.

Quantile regression Estimation of conditional quantiles of Y given covariates; alternative to mean-regression for distributional evidence.

Pre-analysis plan Document committing the analytical specification before data are observed; reference frameworks Olken (2015b) and Christensen and Miguel (2018a).

R-row Single-row pre-registered robustness alternative that varies exactly one design choice from the primary specification; typical universe across closed monthly iterations is R1 (regime-dummy), R2 (sample-cut), R3 (functional-form), R4 (HAC inference).

ESCALATE Clause A Verdict-mapping clause covering positive- $\hat{\beta}$ results with primary p in (0.05, 0.20]; routes the iteration to user adjudication rather than auto-PASS or auto-FAIL.

ESCALATE Clause B Verdict-mapping clause covering near-zero- $\hat{\beta}$ results with high tail asymmetry, operationalised as $|\hat{\beta}/SE| < 0.5$ jointly with $|\text{skew}(\text{residuals})| > 1.0$ or excess kurtosis greater than 3.0; thresholds pinned at spec-authoring time.

SUBSTRATE_T00_NOISY Verdict label triggered when at least three of four R-rows produce sign-flipped composite $\hat{\beta}$ relative to the primary; halts the iteration and requires substrate-level review before any pivot.

On-chain terms

On-chain settlement. Discharge of a financial obligation by execution of a publicly-recorded, programmable rule-set on a public ledger. Traditional-finance analogue: clearinghouse with a public, machine-executable rulebook.

Programmable rule-set. A pre-specified contractual logic, recorded publicly, whose execution is deterministic given observable inputs. Traditional-finance analogue: an ISDA confirmation whose terms are machine-executable rather than discretionarily applied.

Uniswap v3/v4 LP position. A two-asset market-making commitment over a bounded price interval $[P_\ell, P_u]$, whose payoff in the terminal price is piecewise-concave (long-gamma inside the interval, directional outside). Traditional-finance analogue: a passive market-making book with hard inventory limits at the interval bounds.

Perpetual option (on-chain). An option-like instrument with no fixed expiry, whose premium accrues continuously as a streaming funding payment. Traditional-finance analogue: a perpetual swap whose payoff leg is a vanilla option rather than a linear delta.

Panoptic. A deployed protocol that issues perpetual options written against Uniswap v3/v4 LP positions ([Panoptic Labs, 2026](#)). Traditional-finance analogue: an exchange-traded perpetual-future-on-options venue with the option strike encoded as a market-making price interval.

Reserve-backed peg. A stablecoin issuance design in which the pegged claim is collateralised by a reserve portfolio whose composition and valuation are publicly observable. Traditional-finance analogue: a narrow-purpose currency board.

Mento. A deployed reserve-backed stablecoin protocol on the Celo public ledger, maintaining a roster of twelve country-currency-pegged claims as of the 2026-04-27 deployment manifest ([Mento Labs, 2026b](#)). Traditional-finance analogue: a consortium-issued multi-currency narrow bank.

Mento Broker. The routing contract through which Mento exchanges pegged claims against reserve assets at the protocol-determined reference rate ([Mento Labs, 2026b](#)). Traditional-finance analogue: the redemption window of the issuing currency board.

Purchasing-parity stablecoin. A reserve-backed pegged claim denominated in the consumption currency of a target population (COPM, BRLM, KESM, EURM, USDM). A hedge denominated in such a claim settles in the user's consumption currency rather than United States dollars, eliminating residual exchange-rate exposure between hedge settlement and consumption at the unit-of-account level.

Carbon DeFi (excluded). A third-party decentralised exchange that hosts Mento-basket claims as standard fungible tokens. *Carbon has no protocol-level integration with Mento:* the Mento V3 deployment manifest contains zero references to Carbon DeFi or its arbitrage routers. Carbon-Mento trading volume is third-party DEX activity — predominantly Ban-

cor’s own arbitrage routers extracting value from price misalignment between Mento Reserve pricing and secondary-market liquidity — and is *not* Mento Reserve user demand. Any framing of Carbon-Mento volume as a Mento-native demand signal is excluded from this document’s empirical claims.

§B. Cited Literature

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§C. Sample Artifact: Pair D Verdict Header

The Pair D iteration closed PASS on 2026-04-28. The verdict was emitted with a sha256-pinned artifact chain that we reproduce below in condensed form so that the supervisor can verify the audit-trail discipline of the framework. Each artifact is content-hashed; the spec is content-hashed against itself with a sentinel substituted into the `decision_hash` field, which can be recomputed to verify that the document text governing the iteration matches the hash quoted.

- Specification (Pair D simple- β design, version 1.3.1): 964c62cca0be1b9070944b5398fe97886c6d07d37ba7
- Joint estimation panel (`panel_combined.parquet`): 6d7d9e60dad1715ce86e8adb7b3d44ba236d0b0637962
 $N = 134$, window 2015-01-31 through 2026-02-28.
- Primary OLS results (`primary_ols.json`): d4790e743cdec62f1368cab1833e1266cb2da763d7c0931dd732
- Robustness pack (`robustness_pack.json`): 67dd18cfef2584fa6ed9334b1d0314a1a16830faf7c3f3443f0
- Verdict memo (`VERDICT.md`): 1efd0e34d7c1af821c8528a7bc895a63e1dc5e1c289f3b6a1b2d392ba59806c

The specification revision chain that produced the governing hash is: v1 (initial, 2026-04-27) $\rightarrow$ v1.1 (Wave-1 and Wave-2 reviewer defects integrated, 2026-04-27) $\rightarrow$ v1.2 (CORRECTIONS- α , 2026-04-28; window shortened from 2008-01 to 2010-01 after a schema-stability HALT on the DANE empalme nota-técnica coverage) $\rightarrow$ v1.2.1 (three-way-review cleanup, 2026-04-28) $\rightarrow$ v1.3 (CORRECTIONS- α' , 2026-04-28 PM; window shortened from 2010-01 to 2015-01 after a second

HALT under the realization that the 2010–2014 empalme files ship CIU Rev. 3 codes in RAMA4D despite the column header naming) → v1.3.1 (three-way-review cleanup, 2026-04-28 PM late). Each transition recomputed the spec sha256 and re-triggered independent review; the chain is reproducible from the spec text in the project repository. The point of the exhibit is not the specific hash values but the property that every numerical claim in §6.1 is content-hashed back to a specific document text, so that any post-hoc edit to the specification would invalidate the verdict. The compositional reframe noted in §6.1 caveat (v) is metainterpretive — it does not edit the Pair D spec text or the hash chain above; the Pair D verdict and headline numbers stand verbatim under their original spec.

§D. Power-Calculation Transparency Table

This appendix presents whatever per-iteration power derivation is feasible at writing time against the framework’s pre-pinned $MDES = 0.40$ standard-deviation units of Y and $Power_{\min} = 0.80$ at one-sided $\alpha = 0.05$. The sample-size adjustment under autocorrelated residuals follows the classical AR(1) effective-sample-size correction $N_{\text{eff}} = N \cdot (1 - \rho_1)/(1 + \rho_1)$, where ρ_1 is the first-order residual autocorrelation; this is the standard textbook adjustment for the variance of an AR(1) sample mean and is not specifically attributable to a single canonical reference (it predates the HAC literature). Project inference uses Newey-West HAC standard errors (Newey and West, 1987) at fixed lag truncation per §5.4; we report the realized HAC SE where it is recorded in the verdict pack so that the implied effective-sample-size adjustment is auditable from the realized SE itself rather than from a separately-recorded ρ_1 diagnostic. Cells where the derivation cannot be honestly produced from the artifacts available at writing time are flagged *inherited* and cross-referenced to the supervisor-ask in Section 7 (§7.4). We do not fabricate any number we cannot derive from a stable artifact.

The table is the most direct substantive response to the §7.4 ask: it converts the question “is $N_{\min} = 75$ defensible?” from rhetorical to data-supported wherever the data are available, and it surfaces honestly the cells where the data are not available at writing time. The framework does not currently record a single-number residual-autocorrelation diagnostic in the verdict pack of either Pair D, dev-AI Stage-1, or FX-vol-CPI; remediating that artifact gap is itself one of the methodological housekeeping tasks that any forward iteration would inherit, conditional on the supervisor’s guidance.

Iteration	N	$\hat{\rho}_1$ (resid.)	N_{eff}	Implied power at MDES= 0.40	Notes
Pair D	134	inherited	inherited	inherited	$\hat{\rho}_1$ on the primary residuals not re-derived from a stable artifact at writing time; HAC $L = 12$ was used in inference but the residual autocorrelation diagnostic is not recorded as a standalone number in the verdict pack. Realized HAC SE 0.02465 implies the AR(1) adjustment was material. Inherited from Phase-A.0 Rev-5.3.1; re-derivation tabled for §7.4.
dev-AI Stage-1	134	inherited	inherited	inherited (FAIL closed)	Iteration closed FAIL 2026-05-06 with realized HAC($L = 12$) SE 0.0847; the SE is approximately $3.4\times$ the Pair D HAC SE on identical N and lag panel, indicating substantially larger residual variance and/or autocorrelation at the Section-J-narrow stratum. The realized FAIL is sign-flipped, not underpowered, so the post-hoc power question is moot for the verdict; the SE is informative for the next-iteration prior on Section-M (which would inherit a similar autocorrelation structure).
Section M (candidate next iteration)	134 (R2 sample)	not yet estimated as primary	—	not yet estimable as primary	R2 sensitivity arm in dev-AI Stage-1 returned $\hat{\beta} = +0.455$ at $t = +4.73$, one-sided $p = 1.13 \times 10^{-6}$. <i>Post-hoc identified candidate, NOT pre-registered as a primary in any iteration to date.</i> A Section-M iteration would require a fresh spec-pin; the tabulated row is informational and does not graduate the R2 finding.
FX-vol-CPI primary	947	inherited	inherited	inherited	Pre-registered HAC bandwidth $L = 4$ at the weekly cadence; the residual autocorrelation diagnostic is recorded indirectly through the gate-aggregation reconciliation but not as a single-number artifact. Inherited from Phase-A.0 Rev-5.3.1; re-derivation tabled for §7.4.
Phase-A.0 remit-	—	not appli-	not ap-	not appli-	Iteration EXITed at the